

MONGODB ATLAS · HEALTHCARE AI DEMO

Operational data and AI retrieval, one platform.

This demo walks a single synthetic pending healthcare claim through a five-step AI-enrichment loop entirely inside MongoDB Atlas — clinical embeddings, Atlas Vector Search retrieval, hybrid search via **\$rankFusion**, and an AI-generated reviewer rationale — all anchored in the same documents that hold the operational claim data.

The architectural thesis is simple: MongoDB Atlas serves as both the operational system of record and the retrieval foundation for AI workloads. Embeddings live alongside the data they describe; vector and lexical retrieval run as aggregation stages on the same collection; AI output is written back into the original record. No separate vector store, no sync layer, no glue code keeping two systems consistent.

At a glance

- Three engineered scenarios: MRI prior auth, biologic infusion, GLP-1 weight management.
- 30 coverage policies and 71 prior adjudicated claims, all embedded with Voyage AI **voyage-3** (1024 dimensions).
- **\$vectorSearch** and **\$search** indexes live on the same collections; **\$rankFusion** fuses them in one aggregation stage.
- Template-driven rationale — no external LLM API call required.
- Hard operational filters (plan type, state, outcome) are enforced as constraints, not hints, in both vector and lexical pipelines.

THE FIVE-STEP FLOW

From pended claim to written-back rationale.

Every step happens inside MongoDB Atlas. The clinical notes are embedded once and stored in the same document. Both retrieval stages are MongoDB aggregation operators. The AI rationale is written back into the original claim record, preserving a clean audit trail.

Step	What happens	MongoDB role
1	Operational claim record loads — structured fields plus unstructured clinical notes.	Operational database — the source of record.
2	Voyage AI embeds the clinical notes into a 1024-dim vector (voyage-3) and writes it into the same document.	Vector lives next to the data it describes — no separate store.
3	Atlas Vector Search retrieves semantically relevant policies and prior cases, with operational filters enforced.	<code>\$vectorSearch</code> aggregation stage with hard filter clauses.
4	Hybrid retrieval (optional): same query through <code>\$vectorSearch</code> and <code>\$search</code> , fused by <code>\$rankFusion</code> .	Single platform handles both retrieval styles natively.
5	Reviewer-voice rationale generated from retrieved context and written back into the claim record.	AI output stored in the same document as the operational data.

THREE ENGINEERED SCENARIOS

Same architecture, three clinical situations.

Each scenario exercises the same five-step flow against different clinical content, plan type, and pend reason. Scenario B is the intended showcase for hybrid retrieval — the Crohn's contrast case **PCL-2024-BIO-3580** is engineered to surface in hybrid mode but not in pure **\$vectorSearch**.

Scenario A MRI Prior Authorization · CPT 72148 Member James Thornton · 58 M · PPO · OH Pend reason PA-MN-001 — medical-necessity review Clinical focus Lumbar radiculopathy with conservative treatment history; imaging request. Expected outcome APPROVE — conservative therapy documented, criteria met.	Scenario B Specialty Drug Infusion · HCPCS J1745 (Remicade) Member Eleanor K. Vasquez · 44 F · PPO · TX Pend reason MN-DRUG-HCB-002 — high-cost biologic Clinical focus Seropositive RA on infliximab; step-therapy review for \$18,420 claim. Expected outcome APPROVE — step therapy thoroughly documented.	Scenario C GLP-1 Prior Authorization · HCPCS S0148 (Wegovy) Member Patricia A. Reeves · 52 F · PPO · FL Pend reason PA-OBE-GLP1-001 — lifestyle program docs Clinical focus Morbid obesity (BMI 38.2) with T2DM, HTN, OSA; lifestyle documentation gap. Expected outcome ADDITIONAL INFORMATION REQUIRED — qualifies but documentation incomplete.
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WHY THIS ARCHITECTURE

What is distinctive about MongoDB Atlas here.

One platform for operational and AI workloads

Operational claim data, Voyage AI embeddings, retrieval indexes, and AI output all live in the same Atlas cluster — anchored to the same documents. No separate vector database to keep in sync, no data movement to support retrieval.

Hybrid retrieval, natively

\$rankFusion (MongoDB 8.1+) combines a **\$vectorSearch** pipeline and a **\$search** (lexical) pipeline in a single aggregation stage with reciprocal rank fusion. The same hard filters apply to both pipelines, so the comparison stays operationally honest.

Hard filters on semantic retrieval

Plan type, member state, and adjudication outcome are enforced as filters inside the search stage — they are constraints on the results, not hints to a model. This is the difference between useful semantic search and unguarded semantic search in a regulated healthcare context.

AI output written back to the record

Step 5 stores the generated rationale, supporting policy IDs, comparable case IDs, and a status-history entry on the original claim document. Auditability is built in — no separate output store, no ID reconciliation between systems.